APC523 Problem Set 2

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Problem 1

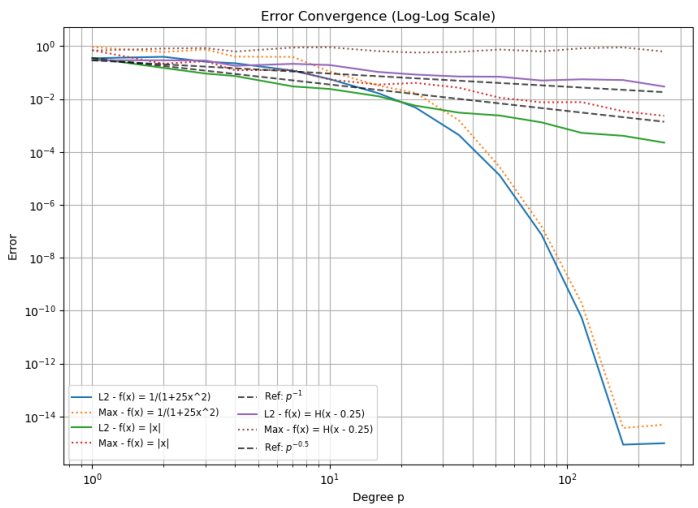

Figure 1: Power Laws for Problem 1

We see power laws for the Heaviside and absolute values functions of 1 and 0.5 respectively. This makes sense as these functions are not continuous and at some point the differentiation falls apart. On the other hand the Runge function has exponential dependency as it is infinitely differentiable.

Problem 2

The power law for x^8 should always be machine precision since the Gauss quadrature guarantees the exact answer, and that is observed in the graph. The power law for $|x - 1/\sqrt{2}|^3$ is observed to be between h^3 and h^4 before hitting machine precision. Although the integration struggles for the Heaviside step function, it shows a dependence around h^1 and the error is very high. The dependence for $f(x) = 1/\sqrt{x}$ is perfectly $h^{0.5}$ but the error is very high and the integrator struggles with this function. These dependencies make sense since at a certain point these functions aren't differentiable.

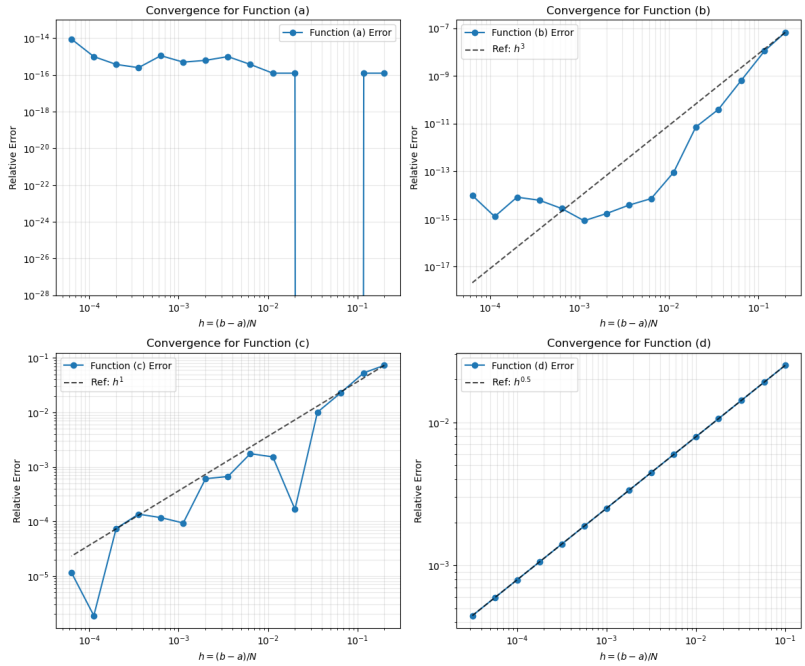

Figure 2: Power Laws

Problem 3

Part 3a

Given these eigen vectors and eigenvalue

$$\mathbf{v}_m = \begin{bmatrix} v_{m,0} \\ v_{m,1} \\ \vdots \\ v_{m,N-1} \end{bmatrix}, \quad \text{where } v_{m,n} = \exp\left(-i\frac{2\pi mn}{N}\right), \quad n, m = 0, \ldots, N-1$$

$$\lambda_m = \alpha - 2\cos\left(\frac{2\pi m}{N}\right).$$

We know that for eigen vector v_m, the relation $T_\alpha v_m = \lambda_m v_m$ needs to hold true. Define

$$u_{m,n} = \alpha * v_{m,n} - v_{m,n-1} - v_{m,n+1}$$

where $u_{m,n}$ is an generic element in the $T_\alpha v_m$ vector. The We know this needs to equal $\lambda_m v_{m,n}$ given the properties of eigenvalues and eigenvectors.

$$v_{m,n-1} = \exp\left(-i\frac{2\pi m(n-1)}{N}\right), v_{m,n+1} = \exp\left(-i\frac{2\pi m(n+1)}{N}\right)$$

Since we know the values of $v_{m,n-1}$ and $v_{m,n+1}$, we can plug into the original equation.

$$u_{m,n} = \alpha * \exp\left(-i\frac{2\pi mn}{N}\right) - \exp\left(-i\frac{2\pi m(n-1)}{N}\right) - \exp\left(-i\frac{2\pi m(n+1)}{N}\right)$$

We can take out like terms and get the following equation.

$$u_{m,n} = \exp\left(-i\frac{2\pi mn}{N}\right) * \left[\alpha - \exp\left(i\frac{2\pi m}{N}\right) - \exp\left(-i\frac{2\pi m}{N}\right)\right]$$

Using Euler's identity, this can be simplified as a cosine.

$$u_{m,n} = \exp\left(-i\frac{2\pi mn}{N}\right) * \left[\alpha - 2\cos\left(\frac{2\pi m}{N}\right)\right]$$

This is clearly $\lambda_m v_{m,n}$, so they must be eigen vectors and eigen values of T_α.

Part 3b

In order for T_α to be positive definite, $\alpha > 2\cos\left(\frac{2\pi m}{N}\right)$, since the eigenvalues need to be positive.

Part 3c

Split T_α into M and N with M being the diagonal component and rearrange for N.

$$\mathbf{M} = \alpha\mathbf{I}, \quad \mathbf{N} = \alpha\mathbf{I} - \mathbf{T}_\alpha \tag{1}$$

Using the formula $\mathbf{G} = \mathbf{M}^{-1}\mathbf{N}$

$$\mathbf{G} = (\alpha\mathbf{I})^{-1}(\alpha\mathbf{I} - \mathbf{T}_\alpha) = \mathbf{I} - \frac{1}{\alpha}\mathbf{T}_\alpha \tag{2}$$

$$g_m = 1 - \frac{\lambda_m}{\alpha} = 1 - \frac{\alpha - 2\cos\left(\frac{2\pi m}{N}\right)}{\alpha} = \frac{2}{\alpha}\cos\left(\frac{2\pi m}{N}\right) \tag{3}$$

$$\rho(\mathbf{G}) = \frac{2}{|\alpha|} \tag{4}$$

For this iterative scheme to converge, it is necessary that $\rho(\mathbf{G}) < 1$, which implies the condition $|\alpha| > 2$.

Part 3d

(1) The convergence rate is the ratio of the error at step k to the error at step $k-1$ (i.e., $e^{(k)}/e^{(k-1)}$).
Observed rate for $\mathbf{v}_1$: ≈ 0.9980267
Observed rate for $\mathbf{v}_2$: ≈ 0.9921147
(2) In Part (c), we derived the eigenvalues of the iteration matrix $\mathbf{G}$ to be:

$$g_m = \frac{2}{\alpha}\cos\left(\frac{2\pi m}{N}\right)$$

Since we are using $\alpha = 2$ and $N = 100$, the convergence rate for a specific eigenvector $\mathbf{v}_m$ is the eigenvalue $|g_m|$.
For $m = 1$: $|g_1| = \cos\left(\frac{2\pi}{100}\right) \approx 0.9980267$
For $m = 2$: $|g_2| = \cos\left(\frac{4\pi}{100}\right) \approx 0.9921147$
The values match exactly.
(3) The overall error is limited by the slowest decaying mode, which corresponds to the largest eigenvalue of the iteration matrix (the spectral radius). The slowest decaying modes are the $m = 1$

and $m = N - 1$ modes. Therefore, the asymptotic convergence rate for an arbitrary vector $\mathbf{y}$ will bottleneck at:

$$\rho = |g_1| = \cos\left(\frac{2\pi}{100}\right) \approx 0.998$$